COMPASS FATE: Sulfide

Sept 2026 Maryland Sample Reruns

2026-10-13

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed
sample_month = "Sept"
sample_year = "2026"
Sample_Month_Year = "202609"
Run_Date = as.Date(c("10/07/2026"), format = "%m/%d/%Y")
Run_by = "Zoe Read"#Instrument user
Script_run_by ="Zoe Read"#Code user

Run_notes=""#any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate.*20261007\\.csv")
(Raw_plates)
```

```
## [1] "Raw Data/COMPASS_H2S_Sept2026_Plate1_20261007.csv"
## [2] "Raw Data/COMPASS_H2S_Sept2026_Plate2_20261007.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

# Metadata path
Raw_Metadata <- "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/FATE_2026_processed.csv"
# rerun_path <- "Processed Data/COMPASS_SynopticCB_H2S_202607_MD_reruns.csv"
```

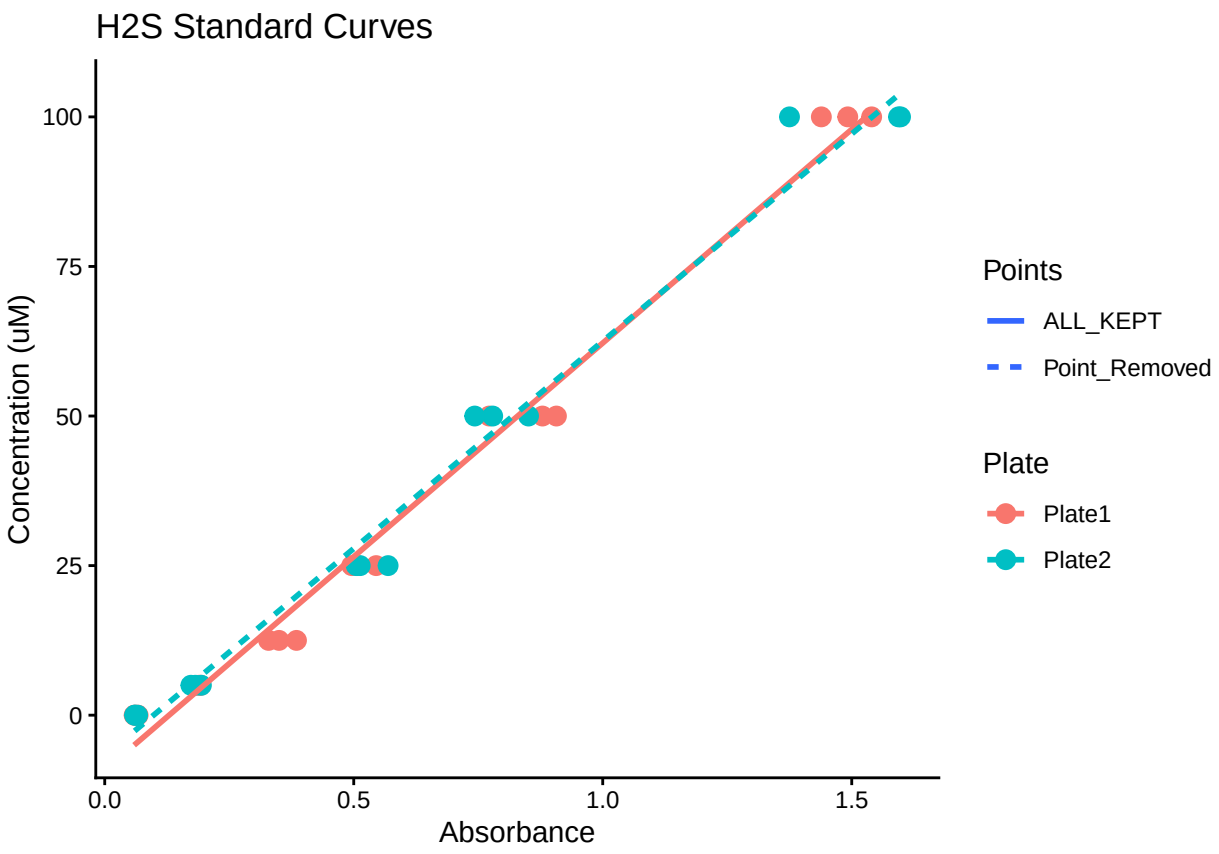
##Read in data

##Remove High CV Stds

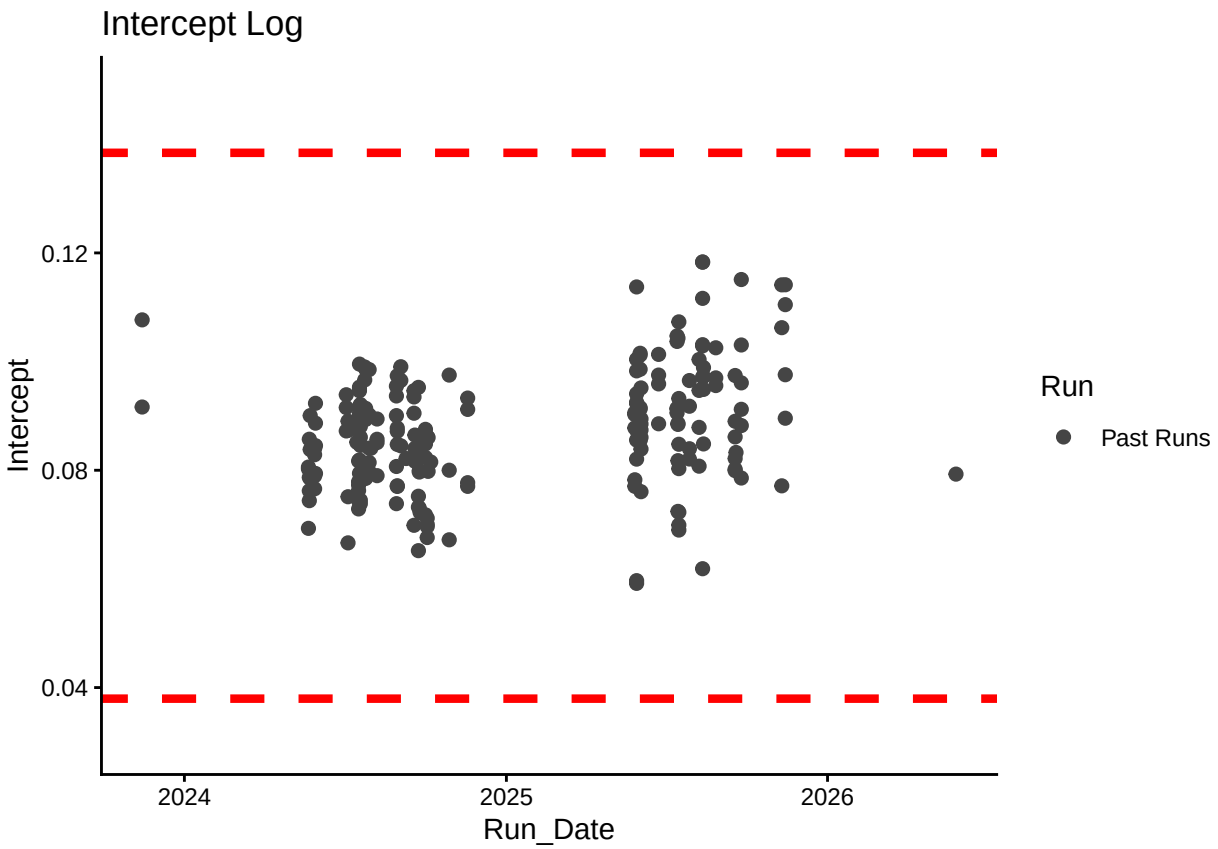
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

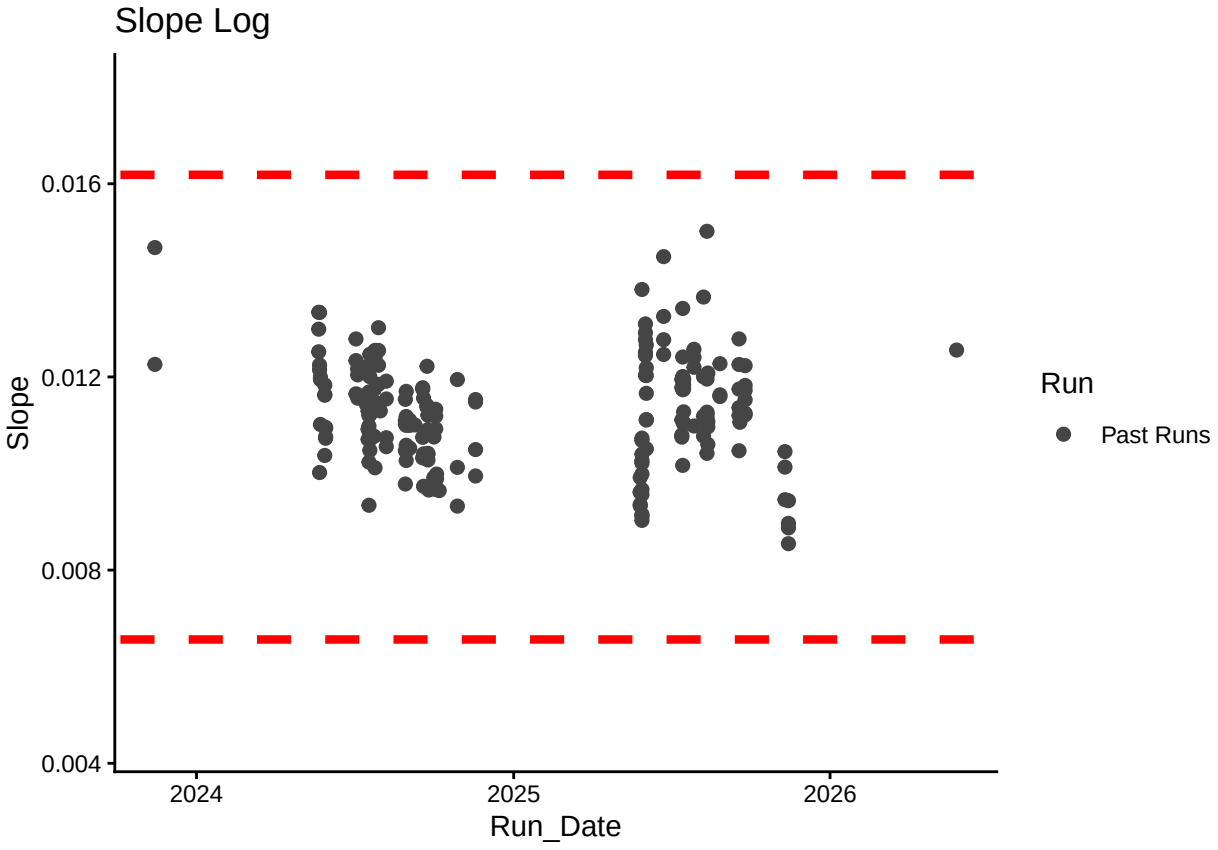
Plot Standard Curves




```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

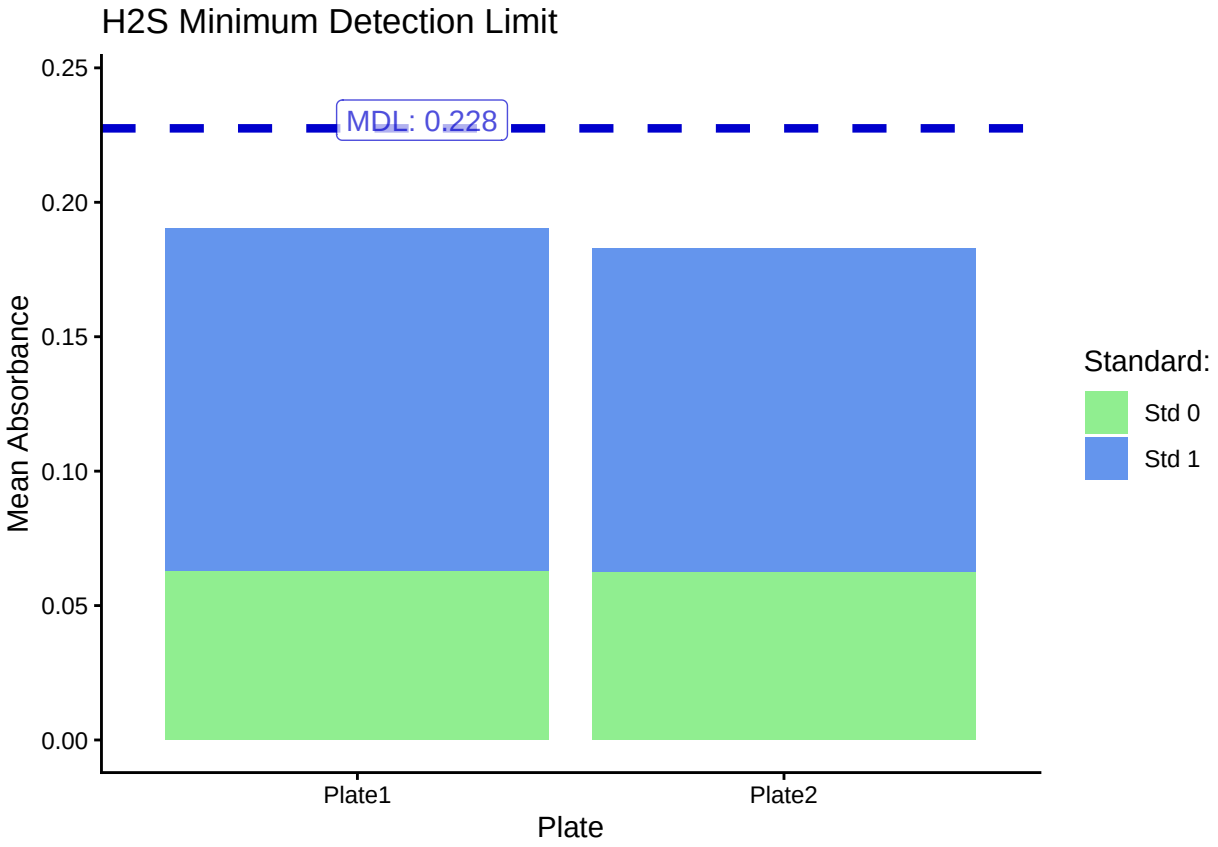


Return Flags For Standard Curve

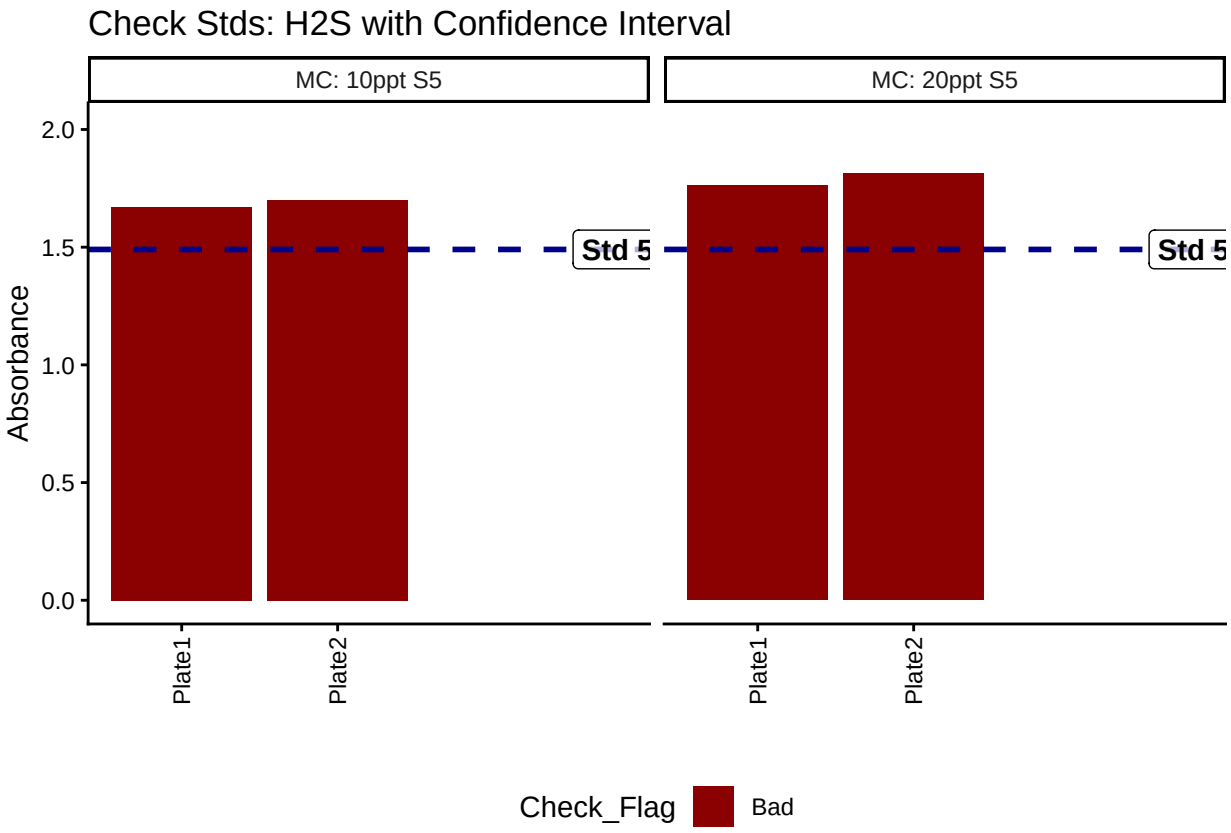
Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate2	REMOVED

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

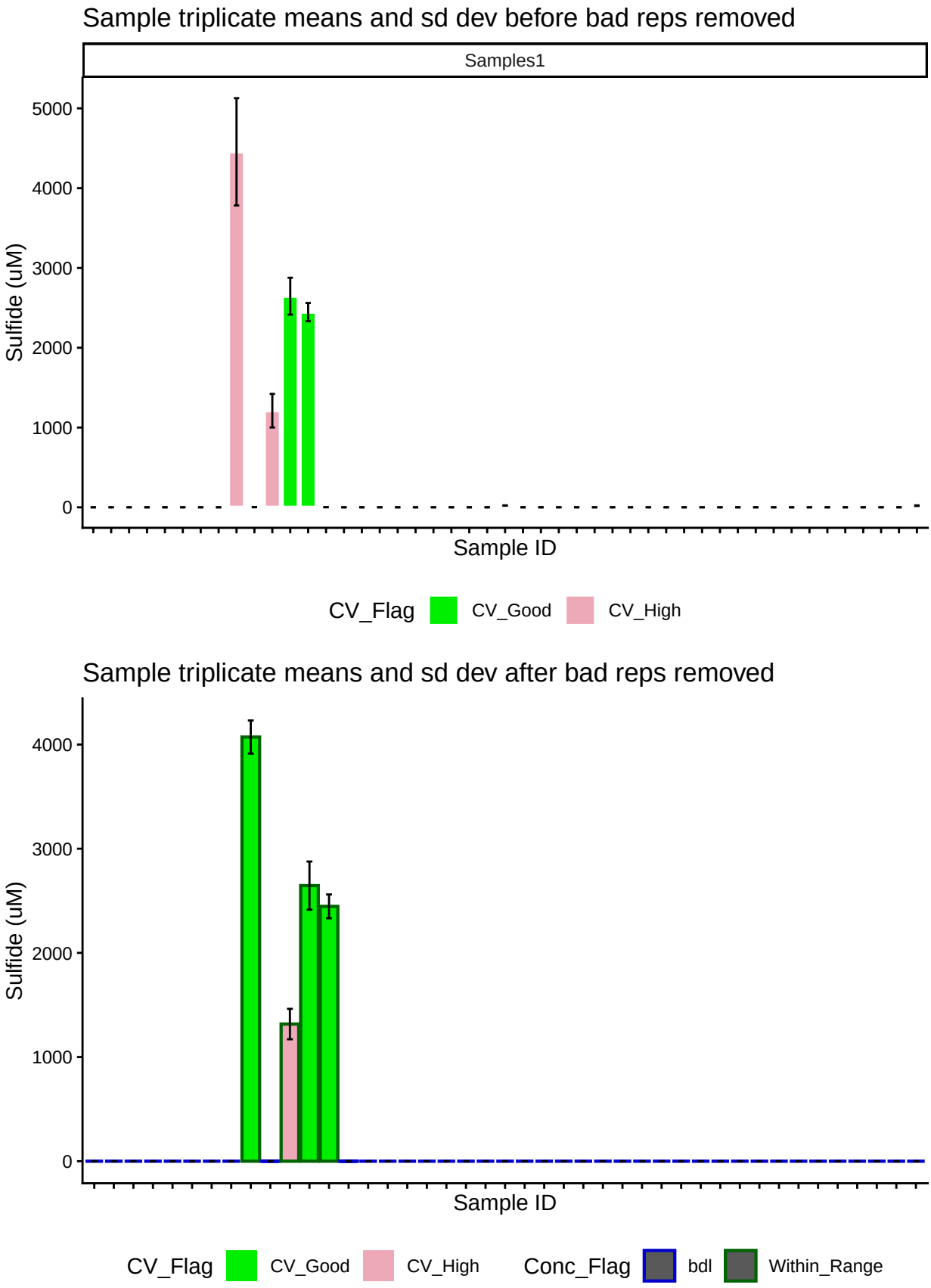
Plate	CHKs_Flag
Plate1	MC: 10ppt S5 Chk_Bad
Plate1	MC: 20ppt S5 Chk_Bad
Plate2	MC: 10ppt S5 Chk_Bad
Plate2	MC: 20ppt S5 Chk_Bad

Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

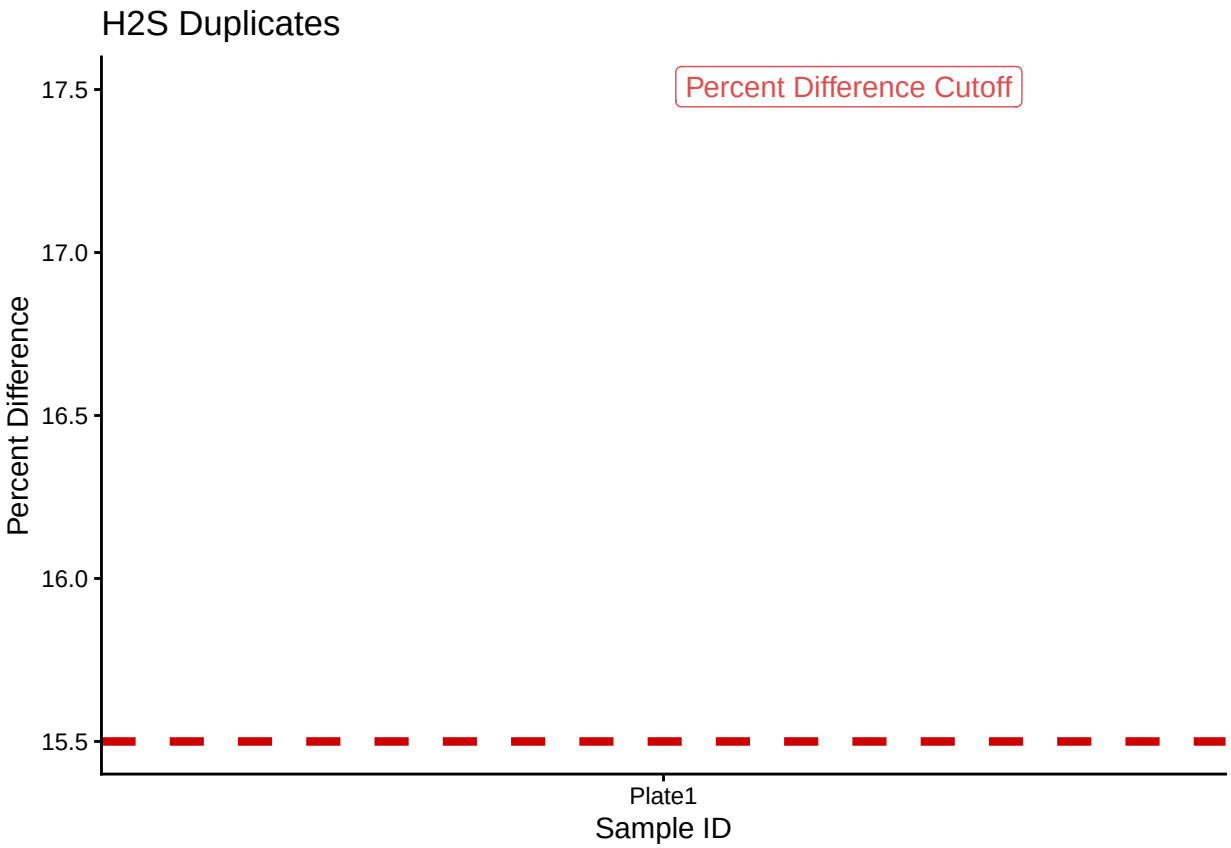
##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC

```
## Warning: No shared levels found between ‘names(values)‘ of the manual scale and the
## data’s fill values.
```

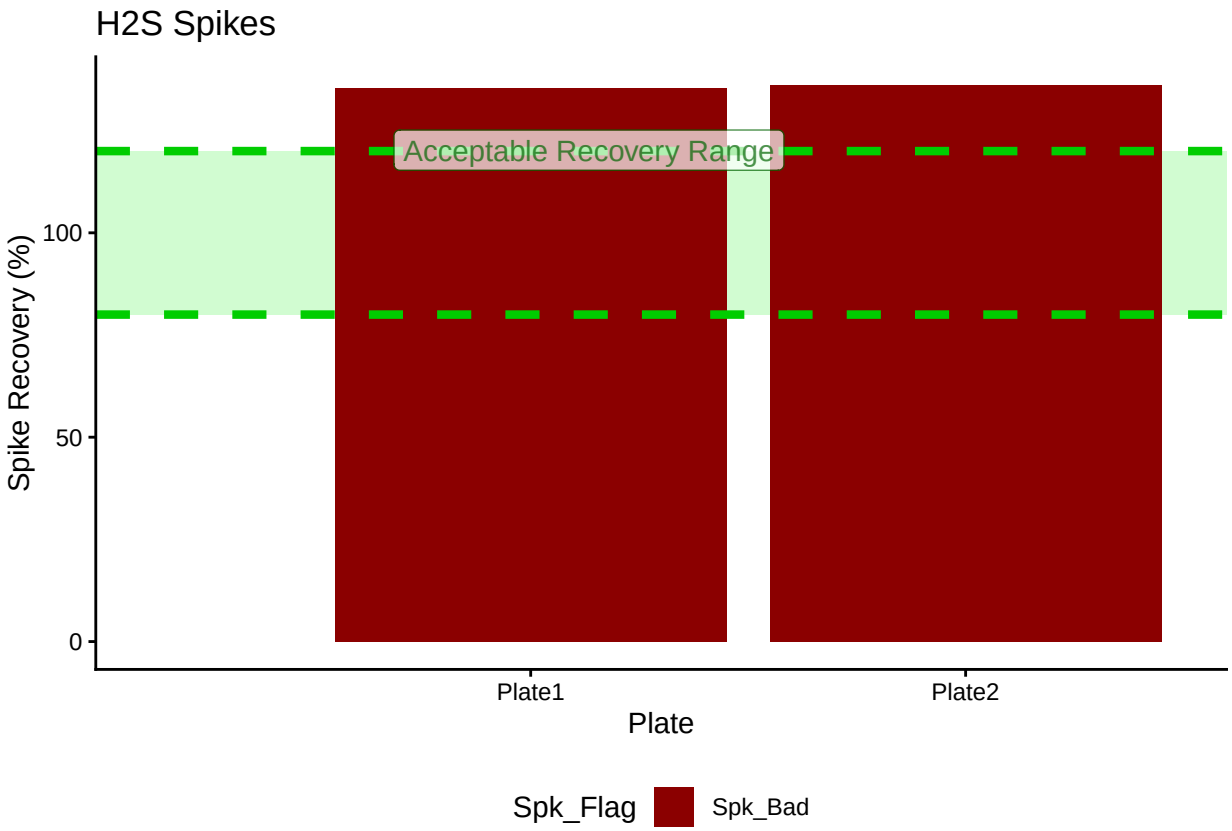


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

##Add Flags

##Remove Synoptic samples and rename TEMPEST samples

```
df_all_clean <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "SWH|GCW|GWI|MSM")) %>%
  filter(!str_detect(Sample_ID, "TMP"))  %>%
  filter(!str_detect(Sample_ID, "dup"))  %>%
  filter(!str_detect(Sample_ID, "spk"))

# ##TMP_C_C6_50_20260618 --> TMP_FW_PW_C6_50cm_SO4_20260618
# Sample_data_IDs <- df_all_clean %>%
#   mutate(Plot = stringr::str_extract(Sample_ID, 'SW|FW|_C_|ESTUARY'),
```

```
#          Plot = gsub("_", "", Plot),
#          Grid = stringr::str_extract(Sample_ID, "B4|C3|C6|D5|E3|F4|F6|H3|H6|I5") ,
#          Sample_Type = stringr::str_extract(Sample_ID, "SOURCE|RAINWATER"),
#          Sample_Type = replace_na(Sample_Type, "PW"),
#          Date = stringr::str_extract(Sample_ID, "[0-9]{8}"),
#          Depth = stringr::str_extract(Sample_ID, "_15_|_30_|_50_"),
#          Depth = gsub("_", "", Depth),
#          Depth = paste0(Depth, "cm")) %>%
#   mutate(Sample_ID = paste("TMP", Plot, "PW", Grid, Depth, "SO4", Date, sep = "_"))
```

##Pull in active porewater inventory "'{#r Pull in active porewater tracking inventory sheet from Google Drive, include=FALSE}

#Run these to pull the TEMPEST porewater metadata into GitHub if not already there

##link directory to google drive folder directory <- "https://drive.google.com/drive/folders/1oEeiU48OULo0UOu0np3l1JhYYE

#filter to necessary files files_pulse <- drive_ls(directory) %>% filter(grepl("TEMPEST_PorewaterInventory_May2022_Presen name))

#download files to local drive lapply(c(files_pulse$name), drive_download, overwrite = TRUE)

sheet_names <- excel_sheets("TEMPEST_PorewaterInventory_May2022_Present_updated.xlsx") sheet_names

raw_metadata_lys <- read_excel("TEMPEST_PorewaterInventory_May2022_Present_updated.xlsx", sheet = "Porewater - Individual", skip = 4)

raw_metadata_sce <- read_excel("TEMPEST_PorewaterInventory_May2022_Present_updated.xlsx", sheet = "Source Water", skip = 4)

#merge metadata raw_metadata_all <- bind_rows(raw_metadata_lys, raw_metadata_sce)

#select SO4/Cl samples raw_metadata <- raw_metadata_all %>% filter(str_detect(Sample_ID, "SO4")) %>% mutate(Sample_ID = gsub("-", "_", Sample_ID)) %>% mutate(Sample_ID = gsub("_Post", " ", Sample_ID))

##Check samples against metadata

'''{#r Check to see if samples run match metadata & merge info, echo = FALSE}

```
Sample_data_IDs$Sample_ID <- toupper(Sample_data_IDs$Sample_ID)
raw_metadata$Sample_ID <- toupper(raw_metadata$Sample_ID)

all_present <- all(Sample_data_IDs$Sample_ID %in% raw_metadata$Sample_ID)

if (all_present) {
  message("***All sample IDs are present in metadata.***")
} else {
  message("***Some sample IDs are missing from metadata.***")

  # Optional: Which ones are missing?
  missing_ids <- setdiff(Sample_data_IDs$Sample_ID, raw_metadata$Sample_ID)
  print(missing_ids)
}

# missing_ids <- data.frame(missing_ids)
#
# ##Export missing IDs to csv
# write.csv(missing_ids, "20250529_COMPASS_H2S_missing_IDs.csv")

#merge metadata with sample run data
merged_data <- Sample_data_IDs %>%
  left_join(raw_metadata, by = c("Sample_ID"))
```

Visualize Data by Plot

“‘{#r Visualize Data, echo=FALSE, warning=FALSE}

#Plot samples to get a first look at concentrations (sanity check) data_plotting <- merged_data

#Order by Zone from Upland to Surface Water data_plotting <- data_plotting %>% mutate(Zone = case_when (str_detect(Sample_ID, “*C*”) ~ “Control”, str_detect(Sample_ID, “*FW*”) ~ “Freshwater”, str_detect(Sample_ID, “*SW*”) ~ “Saltwater”, str_detect(Sample_ID, “*EST*”) ~ “Estuary”)) data_plotting <- data_plotting[order(data_plotting$Zone),]

#group the data for plotting data_plotting <- data_plotting %>% mutate(row_num = factor(row_number())) %>% # create row_num as factor per group ungroup()

#Plot data and change colors based on Zone: viz_H2S_plot <- ggplot(data_plotting, aes(x = row_num, y = Conc, fill = Zone)) + geom_bar(stat = “identity”, position = position_dodge2(preserve = “single”)) + facet_wrap(~ Zone, scales=“free_x”, ncol = 2) + scale_fill_manual(values = c(“Control” = “springgreen2”, “Freshwater” = “cyan2”, “Saltwater” = “violetred2”, “Estuary” = “grey”)) + theme_classic() + theme(axis.text.x = element_blank()) + labs(x = ” “, y =”Sulfide (uM)“, title =”Samples: Sulfide”) + scale_x_discrete(drop = TRUE)+ geom_text(aes(label = ifelse(Conc == 0, “0”, “ “)), vjust = -0.5, size = 2)+ theme(panel.spacing = unit(.05,”lines”), panel.border = element_rect(color = “black”, fill = NA, linewidth = 0.1))

viz_H2S_plot

“‘

##Export Data

END